

Exercise 5: Initial Feasibility Study

A. Technical Feasibility

The project is technically feasible because it can be developed using standard web-development technologies available to the team.

The system requires common technologies such as:

- Database management.
- User authentication.
- Role-based access control.
- File upload for prescriptions.
- Order and inventory management.
- Basic payment simulation/API integration.
- Git/GitHub version control.

None of the core features require specialized hardware.

The system will be divided into manageable software modules, reducing development complexity. More technically complex external integrations, such as government medical databases and production payment processing, are intentionally outside the initial project's scope.

Conclusion: Technically feasible.

B. Economic Feasibility

The academic version of the system can be developed with relatively low cost because most required tools and technologies have free or educational versions.

Expected resources include:

- Development computers already owned by team members.
- GitHub.
- Development frameworks.
- Database software.
- Testing tools.
- Optional hosting.

Real production deployment could require additional costs for hosting, security, backups, payment services, maintenance, and possibly external services, but these expenses are unnecessary for the course prototype.

Conclusion: Economically feasible for the academic project.

C. Operational Feasibility

The proposed system addresses a genuine operational need by giving customers a structured online method to interact with the pharmacy while providing staff with a centralized process for handling prescriptions, orders, inventory, and customers.

Customers benefit from being able to browse and order products remotely.

Staff benefit from having orders and prescriptions organized through one system rather than managing independent manual processes.

The owner benefits from increased control over online products, staff access, inventory, and orders.

Because the system follows activities already associated with normal pharmacy operations rather than creating an entirely unfamiliar workflow, users are likely to understand and accept the system if the interface is simple and reliable.

Conclusion: Operationally Feasible.

D. Schedule Feasibility

The complete commercial version of an online pharmacy could be very large. However, the proposed course version is feasible within the semester because the project scope is limited and development can be divided into increments.

The team will prioritize:

1. User accounts.
2. Medicine catalogue.
3. Ordering.
4. Prescription processing.
5. Staff functionality.
6. Owner functionality.
7. Testing and documentation.

Optional and advanced functionality will only be implemented after the core system is operational.

Using an Incremental Model also supports schedule feasibility because working portions of the software can be completed progressively instead of requiring the entire system to be developed at once.

Conclusion: Schedule is feasible if the scope is controlled.

E. Overall Feasibility

The La Roche Pharmacy Online Prescription and Medicine Management System is considered feasible for the COE420 project.

The required technologies are available to the development team, development costs for a prototype are low, the system addresses real pharmacy and customer needs, and the core functionality can be completed within the course duration if the project scope is carefully controlled.

The main constraints are development time, team experience, security/privacy considerations, integration between modules, and the complexity of prescription and payment processing.

These constraints will be controlled through incremental development, clearly defined roles, continuous testing, limited external integrations, and prioritization of core functionality.